



Deep and Probabilistic Solar Irradiance Forecast at the Arctic Circle



Niklas Erdmann¹, Lars Ø. Bentsen¹, Roy Stenbro², Heine N. Riise², Narada Warakagoda^{1,3}, Paal Engelstad¹

¹ Department of Technology Systems, University of Oslo | ² Institute for Energy Technology (IFE) | ³ Norwegian Defense Research Establishment (FFI)

Purpose

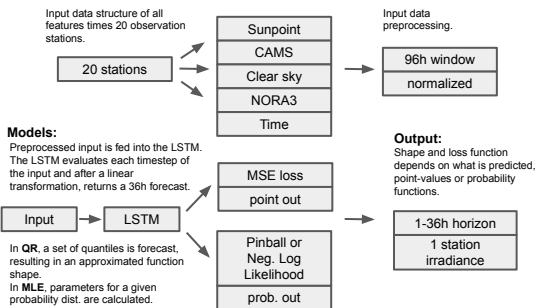
Solar irradiance forecasts can be **dynamic** and **unreliable** due to changing weather conditions. Solar energy producers would benefit from more accurate forecasts in the day-ahead energy market. **Machine learning** has been shown to work in solar irradiance forecasts, but there are less approaches working with **Norwegian data**, or in a **1-36 hours horizon**.

Traditional machine learning forecasts can not tell how reliable their forecasts are, which is important for their useability in the market. Using **probabilistic approaches** on top of a neural network can give a measure of **uncertainty** via a probabilistic output functions. Norway, being close to the **arctic circle**, has harsher differences in solar irradiance depending on **summer or winter**, needing extra consideration.

Method

- > **1-36 hours** forecast of Norwegian solar irradiance using **Long-Short-Term Memory Units (LSTMs)**
- > Uncertainty modelling with **Quantile Regression(QR)** and **Maximum Likelihood Estimation (MLE)** by using the LSTM architecture
- > Comparison of MLE distribution settings: **Gaussian, Johnson's SU, and Weibull**

Data: Sunpoint solar irradiance observations, CAMS-RAD satellite based irradiance, clear sky values, NORA3 NWP weather hindcast features, embedded time

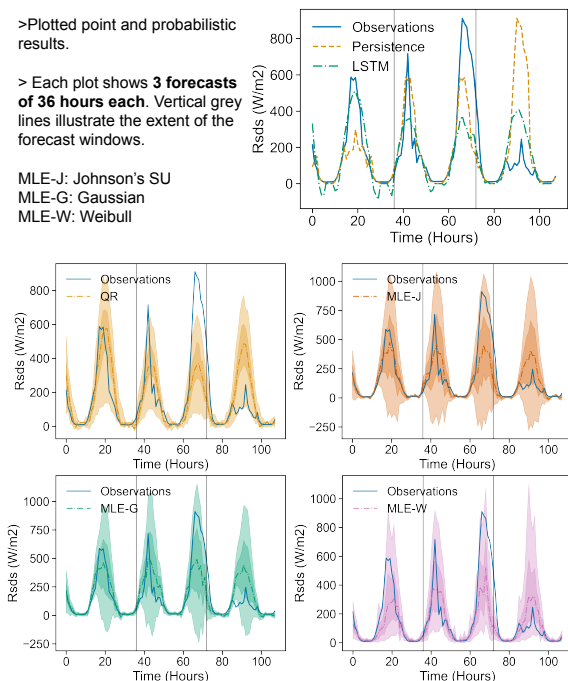


Results

> Plotted point and probabilistic results.

> Each plot shows **3 forecasts of 36 hours each**. Vertical grey lines illustrate the extent of the forecast windows.

MLE-J: Johnson's SU
MLE-G: Gaussian
MLE-W: Weibull



Take away

- > Sunpoint Irradiance forecast with **LSTMs** are **outperforming baseline comparisons**.
- > **Uncertainty** in solar irradiance forecasts **is modelled well** by a neural network driven probabilistic approach such as **QR or MLE**.
- > **MLE beats QR**, but distributions also require careful parameter tuning.

Future Direction:

- > MLE struggles to optimize distribution parameters without manual guidance. Approaches like Variational Inference may perform even better.
- > A point-value LSTM trained on MSE shows no clear edge and worse MAE than MLE-J. MSE loss may need improvement for solar irradiance.
- > Performance of all approaches become more nuanced when taking the strong seasonal differences of the data into account. This needs thorough investigation.

Metric Comparison

	MAE	RMSE	ACE	Q-loss
Persistence	58.376	92.539	-	-
Simple-MLP	69.119	93.156	-	-
LSTM	52.886	77.515	-	-
LSTM-QR	54.062	81.335	0.204	217.429
LSTM-MLE-G	52.832	79.904	0.146	208.424
LSTM-MLE-J	50.783	78.303	0.177	202.324
LSTM-MLE-W	60.155	93.599	0.261	242.909

